unit helps to operate the accelerator rod front and back. This arrangement is made in the fuel injection pump by using a spring arrangement. The cam shaft is present at the bottom of the fuel injection pump. When the cam shaft rotates the cam present in the shaft lifts the plunger unit of the pump. There are provisions made in the barrel to admit fuel and deliver the fuel. The helix present in the plunger in connection with the spil port is designed to suck the fuel and pressurise it. The delivery unit present in the pump at the top has the spring loaded delivery valve and the spring as a single unit. The schematic view of the in line pump can be seen in Fig 9.9.1 and the plunger control rack arrangement can be seen in Fig 9.9.1(a).

Operation:

The in line injection pump of the engine is operated by the timing gear. When the engine is started, the crank shaft is rotated and the power is transferred to the engine cam shaft. From the cam shaft the power goes to the fuel injection pump cam shaft through the timing chain. The plunger present in the fuel injection pump moves up and down due to the cam's rotation. When the plunger is moved down the fuel is sucked inside the barrel of the pump through the inlet port. This happens when the vertical slot is in connection with the spill port and the fuel enters into the pump barrel. To control the amount of fuel delivered to the injector the

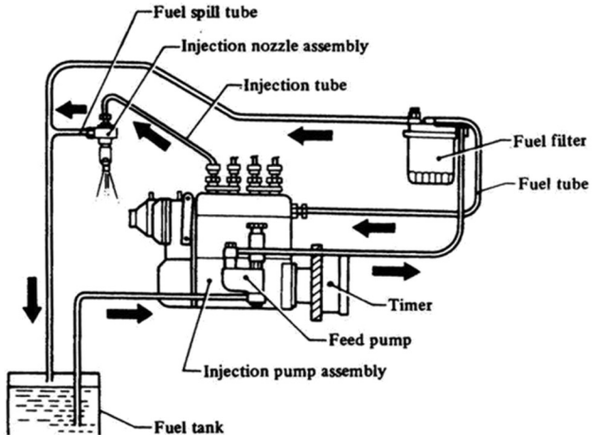


Figure 9.9.1 Schematic View of the In line Pump

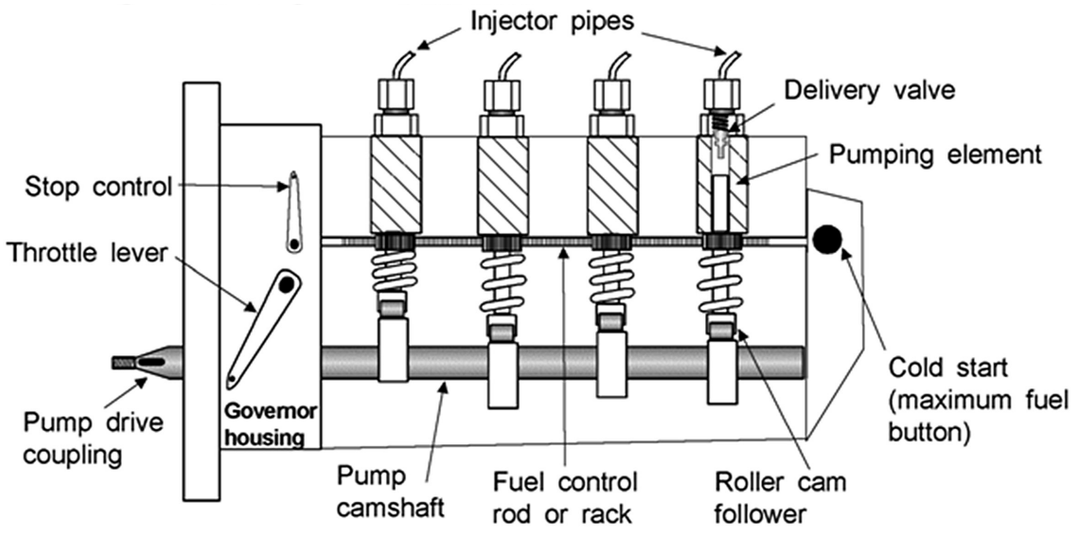


Figure 9.9.1(a) Control Rack and Plunger Arrangement

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